Zichun Zhu

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Email: zhuzichunhogan@gmail.com
Phone: +64-2904526457
LinkedIn: linkedin.com/in/zhuzichun/

Professional summary

Computer Science PhD graduate from The University of Queensland with expertise in machine learning, spatiotemporal data analysis, sequence modelling, and intelligent transportation systems. Most recently led the AI transformation of an ERP software company, building LLM-powered agents, custom models, AI governance tooling, and AI-assisted development workflows. Combines research-level machine learning knowledge with hands-on experience in translating AI methods into real software products and engineering processes.

Education

The University of Queensland, *School of Electrical Engineering and Computer Science* — Brisbane, QLD
Doctor of Philosophy — April 2021 – April 2026

The University of Melbourne — Melbourne, VIC
Master of Science (Computer Science) — July 2018 – June 2020

The University of Melbourne — Melbourne, VIC
Diploma in Informatics — July 2017 – June 2018

The University of Melbourne — Melbourne, VIC
Bachelor of Science (Electrical Systems) — July 2015 – June 2018

Industry experience

Shanghai Little Antelope Software Co., Ltd. — Changzhou, China
AI R&D Contractor — March 2026 – Oct 2026

★ Led the company's transition from a traditional ERP software provider to an AI-oriented, data analytics business.
★ Redesigned and implemented the ERP framework to be AI-compatible and ready for future AI-driven features, including smart factory modules.
★ Built an AI agent integrated into the ERP platform and trained custom models for specific tasks.
★ Built an AI governance platform to manage AI usage across the company and its customers.
★ Introduced an AI-assisted development workflow and AI coding agent tools, and trained teams to use the latest AI tools to improve productivity.

Skills

Languages
Python, Java, SQL, JavaScript, R, C++, MATLAB
AI Engineering
LLM integration, AI agents, custom model training, AI governance, AI-assisted development, coding agents
Machine Learning
PyTorch, TensorFlow, supervised learning, deep learning, sequence-to-sequence models, transformers, contrastive learning, semi-supervised learning, reinforcement learning
Data & Analytics
Data preprocessing, feature engineering, statistical modelling, data visualisation, geospatial data analysis
Tools & Platforms
Linux, VS Code, Git, MySQL, PostgreSQL, MongoDB, AWS (basic), QGIS, Unity3D, Arduino, ROS (Robot Operating System)

Honors and

UQ Research Higher Degree Scholarship with Top-up — April 2021 – April 2026

scholarships

Full doctoral stipend scholarship awarded by The University of Queensland.

Research experience

Bluetooth-Based Trajectory Reconstruction for Urban Mobility Analysis

The University of Queensland — April 2021 – Jan 2026

ARC Linkage Project — Collaboration with Transport and Main Roads (TMR) & Brisbane City Council (BCC)

★ Developed machine learning methods to reconstruct vehicle trajectories from sparse Bluetooth sensor data on urban road networks.

★ Designed and evaluated deep learning, reinforcement learning, contrastive learning, and semi-supervised learning approaches for map-matching and trajectory inference.

★ Built end-to-end pipelines for data preprocessing, model training, and experimental evaluation on real-world transportation datasets.

★ Translated research outcomes into peer-reviewed publications and industry-aligned insights for urban mobility analysis.

Infectious Disease Modelling – Malaria Transmission Dynamics

University of Melbourne — May 2019 – July 2020

★ Developed a population-level simulation model incorporating within-host parasite dynamics to investigate malaria transmission.

★ Applied stochastic simulation and parameter calibration to improve biological realism and model robustness, and to analyse how within-host interactions influence epidemic patterns and transmission outcomes.

Project experience

Quant Trading Bot

Personal — 2026 – Present

★ Built an autonomous research-to-trading pipeline that uses an LLM agent to generate and validate trading strategies before live deployment.

★ Self-hosted a backtesting/execution engine and web control panel to explore automated market analysis and algorithmic trading.

RoboMaster - Simulation Team

University of Melbourne — Sep 2019 – May 2020

★ Developed simulation components in Unity3D and C# to support autonomous robot training and testing.

★ Collaborated in a multidisciplinary team on robotics simulation and AI challenge development.

Autonomous Pacman

University of Melbourne — 2019

★ Built an autonomous game-playing agent in Python using search and reinforcement learning methods, including Monte Carlo Tree Search and Q-learning.

Twitter Analysis

University of Melbourne — 2019

★ Developed a scalable pipeline for collecting and analysing Twitter data on HPC infrastructure using CouchDB, MapReduce, and NLP methods.

FPGA Computer and Calculator

University of Melbourne — 2019

★ Designed and implemented an embedded system on an FPGA board, including a custom CPU, ROM, and peripheral interfaces using Verilog.

★ Developed a stack-based Reverse Polish Notation (RPN) calculator supporting signed 8-bit addition and multiplication, with overflow detection and LED status indicators.
★ Integrated concepts from digital system design, microprocessor architecture, finite state machines, and hardware-software co-design in a complete hardware implementation.

Publications

- **Map-matching on wireless traffic sensor data with a sequence-to-sequence model**. Zichun Zhu, Dan He, Wen Hua, Jiwon Kim & Hua Shi. 24th IEEE International Conference on Mobile Data Management (MDM), 2023.
- **CLMM: Uncertainty-aware Map-Matching for Bluetooth Data through Contrastive Learning**. Zichun Zhu, Fengmei Jin, Wen Hua & Jiwon Kim. 35th Australasian Database Conference (ADC), 2024.
- **A Study on Wireless Traffic Sensor Data Map-Matching**. Zichun Zhu, Wen Hua & Jiwon Kim. PhD Thesis.

Professional development

Conference Presenter
Singapore — 3 July 2023 - 6 July 2023
★ *IEEE International Conference on Mobile Data Management (MDM)*
Gold Coast & Tokyo — 16 December 2024 - 18 December 2024
★ *Australasian Database Conference (ADC)*

Referees

Prof. Wen Hua
Associate Professor — Department of Data Science and Artificial Intelligence
The Hong Kong Polytechnic University
wency.hua@polyu.edu.hk

Dr. Dan He
Data Scientist — Strategy Directorate
Queensland Fire Department
doris.he@qfes.qld.gov.au